

DESIGN SPEC • QUIKBOLUS PHASE 2 • DRAFT V0.1

QuikBolus Phase 2 — Mold Workflow + Printer Profiles

Spec version: 0.1 (draft) **Date:** 2026-05-24 **Author:** Radiant Medical Physics Consulting / Claude Opus 4.7
Status: Design draft — ready for scoping; not yet committed to a release window. **Scope:** Two related additions to the QuikBolus web app — (1) a mold-design mode for casting Flexibolus or similar pourable tissue-equivalent silicones, and (2) a per-printer profile system that surfaces recommended settings and optionally bakes them into a 3MF export.

1. Executive summary

QuikBolus today produces one output: a watertight STL of a patient-anatomy bolus. Operators take it to a slicer, dial in settings, and print on whichever 3D printer they happen to own. Two gaps cause real friction:

1. The **direct-print TPU 95A bolus** isn't the right answer for every clinical case. For thick bolus, dose-critical electron treatments, or chest-wall coverage where print times exceed 36 hours, a **printed rigid mold + cast Flexibolus** workflow is faster, more water-equivalent, and reusable. This workflow currently lives entirely in the operator's head and CAD tools.
2. **Print settings vary by printer** but the operator gets no guidance from QuikBolus. Wrong infill, wrong supports, wrong orientation = clinical failure modes (sparse infill → bad dose; wrong orientation → support scars on the patient-contact surface).

This spec addresses both. **Part 1** designs an interactive Mold mode inside QuikBolus. **Part 2** designs a Target-Printer profile system that surfaces recommended settings per printer and optionally writes a 3MF with the profile baked in.

2. Motivation

PAIN TODAY	SPEC ADDRESSES IT VIA
Operator picks 15% gyroid infill because that's the slicer default → bolus has wrong dose	Per-printer profile UI shows correct settings (100% infill, etc.) directly in QuikBolus
Chest-wall bolus print takes 50+ hours of TPU; clinic resists; operator improvises	Mold mode lets operator print a rigid mold in ~10 hours and cast Flexibolus in 4 hours of cure time
Trim-base-plate workflow is undocumented; operators either skip the base plate or leave it on without re-planning	Mold mode formalizes the "keep vs trim" choice and documents both paths in the output
Operator picks supports manually, often gets them wrong, lands support scars on patient-contact surface	Profile system encodes "orient anatomy-down, tree-organic supports, Z gap 0.25 mm" per printer
Operator on a non-Bambu printer (Prusa, Raise3D, Ultimaker) gets no guidance	Profile system covers ~12 common printers with sensible defaults

3. Scope

In scope

- New "Mold for casting" output mode in QuikBolus
- Interactive parting-plane editor with live 3D preview
- All variables enumerated in §4
- Per-printer profile picker for both direct-print and mold-print outputs
- Recommended-settings panel displayed in the UI
- 3MF export with embedded Bambu / Prusa / Orca profiles for top printers
- STL + recommendations-text-file export for everything else
- Auto-validation checks (undercut detection, volume mismatch, fit check)
- Patient-record entry for both workflows
- Generated per-case operator instruction sheet (PDF)

Out of scope

- Direct printer control (sending G-code over LAN) — covered by a separate Bambu/Orca integration spec
 - New material types beyond TPU 95A, Flexibolus, silicone variants
 - Resin-printed molds (different workflow, different vendor ecosystem)
 - Automatic anatomical-orientation detection from DICOM (parting plane stays manual; "suggest" only)
 - Slicing inside QuikBolus (still farmed out to Bambu Studio / Orca / Prusa / Cura — QuikBolus produces files those slicers consume)
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4. Part 1 — Mold-design workflow

A new output mode in QuikBolus. After the operator generates the bolus from DICOM, they pick **Output mode: Direct print · Mold for casting · Both**. If they pick Mold (or Both), a new mold-design view appears with the variables below.

4.1 Casting material — declared first

Drives almost every downstream default.

CHOICE	NOTES
Flexibolus	Most common; shrinkage ~1%; room-temp pour; cure ~4 h; default
Silicone (Smooth-On Ecoflex, Dragon Skin)	Hot-pour; mold needs PETG over PLA; different wall-thickness defaults
Gelatin / agarose	Test casts only; not for clinical use
Custom	Operator enters density, cure temperature, shrinkage manually

This choice is **first** in the UI because it locks in defaults for the rest.

4.2 Mold construction approach

VARIABLE	CHOICES
Number of parts	Single open-top mold · Two-part split mold (default) · Multi-part (rare)
Mold print material	PETG (default — clear, smooth, durable) · PLA (cheapest) · ABS (heat-resistant for hot-pour silicone) · Resin (smoothest interior)
Reuse target	Single-cast disposable · 5-10 casts (default) · 100+ (production)

Reuse target drives wall thickness in §4.4.

4.3 Parting plane — the critical interactive choice

The single biggest design decision. Needs **live 3D preview** with the split shown in two colors and undercut highlighting in red.

VARIABLE	CHOICES
Parting direction	Horizontal (anatomy / air-side split — default) · Vertical (left/right) · Custom angle
Plane height	Interactive slider — drag to position the split through the bolus
Plane type	Flat (default) · Stepped (for multi-undercut anatomy) · Curved (rare)
Auto vs manual	"Suggest optimal plane" button (minimizes undercuts); operator confirms
Undercut detection	Live overlay highlighting non-releasing regions in red

4.4 Mold geometry

VARIABLE	CHOICES / RANGE
Wall thickness	3 mm (thin/fast) · 5 mm (default) · 8 mm (reusable production)
Floor thickness	Independent of wall; 4-6 mm typical
Outside shape	Rectangular bounding box · Contoured to bolus envelope (default — less material) · Custom
Cavity scaling factor	1.005-1.02 (compensates for Flexibolus shrinkage) — default per casting material from §4.1
Air space above cavity	5-15 mm headroom above bolus for mixing/expansion

4.5 Pour features

Sprue (pour-entry channel)

VARIABLE	CHOICES
Location	Click-to-place on mold surface · Auto at high point in pour orientation
Quantity	1 (small bolus) · 2 (large / chest-wall — accelerates filling)
Diameter	3 mm · 6 mm (default) · 10 mm
Profile	Straight cylinder · Tapered funnel (default — easier to pour)
Extension above mold	0 mm (flush) · 5 mm (default) · 10 mm

Vents (air escape)

VARIABLE	CHOICES
Locations	Click-to-place multiple · Auto-place at every local high point on cavity surface
Diameter	1-2 mm (just enough for air, not material)
Quantity	Auto-suggested by cavity geometry

Risers (gravity reservoirs)

Skip for room-temp Flexibolus. Needed only for hot-pour silicones that shrink on cooling. Out of scope for v1.

4.6 Mold-half registration

VARIABLE	CHOICES
Type	Pegs & sockets (default) · Dovetail interlock · Magnetic inserts (production) · None
Quantity	3 · 4 (default, rectangular layout) · 6+ (large molds)
Peg diameter	4 mm · 6 mm (default) · 8 mm
Peg depth	5 mm · 8 mm (default) · 12 mm
Fit clearance	Tight 0.0 mm · Slip 0.15 mm (default) · Loose 0.3 mm
Placement	Auto at corners (default) · Manual click-to-place

4.7 Clamping / closure

VARIABLE	CHOICES
External clamping ribs	None · 2 sides · 4 sides (default)
Strap channels (for rubber bands)	None (default) · Default groove pattern
Screw bosses (heat-set inserts)	None (advanced; requires inserts; out of scope for v1)
Recommended closing method (informational)	Spring clamps · Rubber bands · Tape · Weight

4.8 Demolding aids

VARIABLE	CHOICES
Draft angle on vertical walls	0° · 1° (default — eases release) · 2° · 3°
Pry-point notches	None (default) · 2 (opposite sides) · 4 (corners)
Surface finish recommendation (informational)	Standard · Smooth (polish interior) · Vapor-smooth (ABS only)

4.9 Identification & traceability

VARIABLE	CHOICES
Patient identifier	None (default) · Initials only · Anonymous ID (auto-hash) · MRN (de-identified)
Location of text	Inside mold (visible only on cast) · Outside mold (default — visible on mold) · Both
Date code	None · YYYYMMDD (default) · Free text
Site / fraction info	None (default) · Treatment site abbreviation · Fraction count
Embossed depth	0.4 mm subtle · 0.8 mm (default) · 1.2 mm obvious
Recessed vs raised	Recessed in mold = raised on cast (default, more readable on cast)

4.10 Print-orientation hints for the mold itself

QuikBolus emits these as recommendations alongside the STL/3MF export. The slicer ultimately handles them.

VARIABLE	RECOMMENDED FOR RIGID MOLD
Print orientation per half	Parting plane DOWN on build plate (cavity opens upward)
Support type	Normal supports (rigid PLA/PETG, not the tree-organic used for TPU)
Support locations	Only at sprue/vent overhangs
Layer height	0.16 mm (smoother cavity interior — transfers to cast surface)
Wall count	4 (mold walls take pour pressure)
Infill	25-40% (rigid material; mold not dose-relevant; some structural strength)

4.11 Cast-process documentation output (not geometry)

Bundled into the per-case operator instruction sheet generated by QuikBolus:

- Flexibolus volume needed (from cavity volume; in mL and grams)
- Mix ratio (per Flexibolus product spec)
- Pour temperature (room temp for Flexibolus)
- Cure time at room temperature
- Recommended mold release (Smooth-On Universal or equivalent)
- Degassing recommendation (vacuum chamber 60 s before pour)
- Demold timing
- Expected cast cost (Flexibolus per-mL price × volume)

4.12 Validation / QA — automatic pre-export checks

QuikBolus runs these before allowing export. Failures show in-context with 3D highlights.

CHECK	FAILURE ACTION
Cavity volume vs design-bolus volume	Warn if >5% mismatch
Wall thickness everywhere ≥ minimum (from §4.4)	Block export ; show problem regions
Undercuts from chosen parting plane	Warn + highlight in 3D preview
Sprue at lowest pour-orientation point	Warn if not
≥1 vent at highest pour-orientation point	Warn if not
Total mold print volume vs target printer build volume (from Part 2)	Block export if doesn't fit; suggest split or larger printer
Estimated Flexibolus cost	Display (operator may adjust cavity offset)

4.13 Output formats

VARIABLE	CHOICES
Per-half output	Two STL files (default) · Single STL with halves pre-positioned · 3MF with both halves as separate objects
Embedded print profile	None · Bambu Studio · Orcaslicer · Prusaslicer · Cura
Generated instructions PDF	Per-case operator instruction sheet with patient ID, casting parameters, expected timing
Patient record entry	Auto-log the mold design + cast parameters to the patient's QuikBolus case

5. Part 2 — Per-printer profile system

A **Target Printer** dropdown in QuikBolus drives recommended settings shown in the UI and (optionally) baked into a 3MF export. Applies to both direct-print bolus and mold-print outputs.

5.1 Printers supported

Top of the clinical 3D-printing market. Other printers fall back to "Generic / Custom" with universal recommendations.

VENDOR	MODELS
Bambu Lab	P1S, P1P, X1-Carbon, X1E, A1, A1 mini, H2D
Prusa Research	MK4 / MK4S, MK3S+, XL (multi-tool), Mini+
Raise3D	Pro3, Pro2, E2
Ultimaker	S3, S5, S7
Markforged	Mark Two
Generic / Custom	Universal recipe + manual override

5.2 Per-printer settings displayed in UI

When operator picks a printer, the recommended-settings panel renders:

TARGET PRINTER: Bambu Lab P1S

Material: TPU 95A
Print bed: Textured PEI
AMS: BYPASS – external spool
Nozzle: 0.4 mm hardened steel
Build volume fit: ✓ (235×235×235; bolus 145×210×8)

Profile settings:
Layer height 0.20 mm
Wall loops 3
Top/bottom shells 4
Infill 100% rectilinear
Nozzle temp 240 °C
Bed temp 55 °C
Outer wall speed 25 mm/s
...

Supports:
Type Tree (organic)
Z gap 0.25 mm
Threshold angle 30°

Estimated print:
Time ~6 h 40 min
Filament use 62 g (~\$2.50)

[Download .3mf with profile baked]
[Download STL only]
[Open in Bambu Studio (if installed)]

5.3 Settings universal vs printer-specific

UNIVERSAL (MATERIAL + CLINICAL)	PRINTER-SPECIFIC
Material choice (TPU 95A)	Nozzle temperature ($\pm 5^{\circ}\text{C}$ between brands)
Infill % (100% for clinical)	Bed temperature (depends on plate surface)
Wall / shell counts	Print-speed ceiling (printer motion stiffness)
Z-gap for supports	Pressure / linear advance (printer firmware)
Brim size	Retraction tuning (direct-drive vs Bowden)
Patient-contact orientation	Build volume / fit check
Layer height	AMS / MMU warning (Bambu / Prusa multi-material)

5.4 Bambu Lab — P1S / P1P / X1-Carbon / X1E

TPU settings are common across the P1 / X1 series. Only enclosure status differs (P1P is open — fine for TPU but watch for drafts).

SETTING	P1S	P1P	X1C	X1E
Enclosure	Yes	No (open)	Yes (clear)	Yes (enterprise)
Bed	Textured PEI	Textured PEI	Textured PEI / Smooth	Textured PEI
AMS	Yes (BYPASS for TPU)	Optional	Yes (BYPASS)	Yes (BYPASS)
Nozzle temp (TPU 95A)	240 °C	240 °C	240 °C	240 °C
Bed temp	55 °C	55 °C	55 °C	55 °C
Recommended speed	25-50 mm/s	25-50	25-50	25-50
Build volume	256 ³	256 ³	256 ³	256 ³

5.5 Bambu Lab — A1 / A1 mini

SETTING	A1	A1 MINI
Enclosure	None (open bed-slinger)	None
Build volume	256×256×256	180×180×180
TPU support	Officially supported; speeds lower than P1/X1	Same
Nozzle temp	235-240 °C	235-240 °C
Bed temp	55 °C	55 °C
Recommended speed	20-40 mm/s	20-40 mm/s
Notes	Lower acceleration than P1 (bed-slinger vibration)	Mini volume limits to small H&N patches only

5.6 Prusa — MK4 / MK4S / MK3S+

SETTING	MK4 / MK4S	MK3S+
Extruder	Direct drive (Nextruder)	Direct drive (E3D)
Build volume	250×210×220	250×210×210
Bed	Textured / smooth steel sheet	Same
TPU profile	PrusaSlicer "Generic Flex 98A" — adjust for your TPU	Same
Nozzle temp	230-235 °C (slightly cooler than Bambu — different extruder)	230-235 °C
Bed temp	50-55 °C	50-55 °C
Recommended speed	25-45 mm/s	25-40 mm/s
Linear advance	0.04-0.05	0.04-0.05
MMU3 / MMU2S	BYPASS for TPU	Same

5.7 Raise3D — Pro3 / Pro2

SETTING	PRO3 / PRO2
Extruder	Dual direct drive
Build volume	300×300×300 (single head; dual mode reduces)
Bed	BuildTak / glass options
Notes	Industrial-grade; heavier, slower; very reliable for large prints
Nozzle temp	230-240 °C
Bed temp	55-60 °C
Recommended speed	30-50 mm/s
Enclosure	Full (good for thermal stability on big prints)
TPU profile	ideaMaker flex profiles; large build volume = ideal for full chest-wall bolus

5.8 Ultimaker — S3 / S5 / S7

SETTING	S3 / S5 / S7
Extruder	Print Core (swappable); AA0.4 / BB0.4 for TPU
Build volume	S3: 230×190×200 · S5: 330×240×300 · S7: 330×240×300
Bed	Glass with adhesive (PrintCore-specific)
Nozzle temp	225-235 °C
Bed temp	60-70 °C (higher than Bambu/Prusa due to glass)
Recommended speed	25-40 mm/s
Cura profile	"Ultimaker TPU 95A" — adjust infill to 100%
BB extruder for PVA supports	Useful for mold workflow (PVA washes away clean)

5.9 Markforged — Mark Two

SETTING	MARK TWO
Build volume	320×132×154
TPU support	Limited — Markforged ships Onyx (nylon + CF) primarily
Use case	Better for the RIGID MOLD half of the mold-and-cast workflow (Onyx mold is very durable for production), not for the bolus itself
Recommendation	Pair with separate TPU/Flexibolus workflow

5.10 Generic / Custom

For everything else, expose the universal recipe and let the operator dial it in.

SETTING	RECOMMENDED STARTING POINT
Material	TPU 95A
Layer height	0.20 mm
Wall loops	3
Top/bottom shells	4
Infill	100% rectilinear
Nozzle temp	235 °C (tune $\pm 5^{\circ}\text{C}$)
Bed temp	55 °C (tune for plate)
Speed	30 mm/s outer wall, 50 mm/s infill
Retraction	0.5 mm @ 30 mm/s
Extruder	Direct-drive strongly preferred (Bowden + TPU is painful)
Enclosure	Optional for TPU
Build volume	Must fit bolus — QuikBolus fit-checks before export

6. UX flow

6.1 Direct-print workflow (existing + profile system addition)

1. Operator generates bolus from DICOM (as today)
2. Operator picks **Output mode: Direct print** (new dropdown)
3. Operator picks **Target printer** from list (new dropdown)
4. QuikBolus shows the recommended-settings panel from §5.2
5. Build-volume fit check runs; warns if bolus doesn't fit
6. Operator clicks **Download .3mf** (with profile) or **Download STL** or **Open in slicer** (if installed)
7. Patient-record entry auto-logged with chosen printer + settings

6.2 Mold workflow (new)

1. Operator generates bolus from DICOM (as today)
2. Operator picks **Output mode: Mold for casting**
3. Operator picks **Casting material** (§4.1) — drives downstream defaults
4. **Mold-design view** opens with live 3D preview (Three.js or similar)
5. Operator iterates through parting plane (§4.3), geometry (§4.4), pour features (§4.5), registration (§4.6) — each change updates the 3D preview

6. Validation panel (§4.12) shows running warnings + blocks
7. Operator picks **Target printer** (§5) for the mold
8. Operator clicks **Export molds** — produces: - Two STLs (one per half) OR 3MF with both halves - Per-case PDF instruction sheet (§4.11) - Patient-record entry
9. Operator prints the molds, pours Flexibolus per the instruction sheet, demolds, applies to patient

6.3 Both modes

Operator can pick **Output mode: Both** and export everything — useful when they want options or when the planner hasn't decided yet.

7. Open design questions

These deserve discussion before scoping the work.

#	QUESTION	MY CURRENT LEAN
1	Auto-detect best parting plane, or manual only?	Manual with "Auto-suggest" button. Auto-only is appealing but algorithms that minimize undercut count can choose planes that are hard to print or demold. Operator always confirms.
2	Should the per-printer 3MF profiles be live (bake actual slicer settings into the .3mf) or text-only (recommendations on screen + plain STL)?	Both. Live for top 3 printers (Bambu P1S, X1C, Prusa MK4) because that's where the convenience pays off. Text-only for the long tail because slicer-format changes break baked profiles.
3	Live 3D preview in the browser — Three.js or a heavier engine?	Three.js. Three.js handles boolean previews, mesh visualization, and interactive sliders comfortably. STL parsing libraries exist. Heavier engines (Babylon, model-viewer) are overkill.
4	Mold geometry computation in browser or server-side?	Server-side. Boolean subtract on real anatomical meshes is heavy; do it in Python with trimesh or pymeshlab and return STL/mesh to the browser. Browser shows the preview, server does the math.
5	Embedded patient identifiers in the mold — opt-in or default-off?	Default-off. PHI concerns. Operator must explicitly choose to embed any identifier, and the field shows what will go in.
6	Cast volume + cost estimate — show in mL only, or convert to vendor SKUs (Flexibolus 50 grams = X dollars)?	mL + grams only for v1. SKU pricing changes; let the operator do the math. v2 could add a cost-lookup table maintained by the clinic.
7	Should QuikBolus handle slicing internally, or always export to external slicer?	Always external. Slicing is a vast, fast-moving space (PrusaSlicer / Bambu Studio / Orca / Cura). QuikBolus stays focused on the geometry generation; slicing is someone else's job. The 3MF profile bake is the integration point.

8. Implementation phasing

Suggested release sequence. Each phase ships independently.

Phase 2.1 — Printer profile system (direct-print)

- Target Printer dropdown
- Recommended-settings panel
- Build-volume fit check
- Profile JSON per printer (text-only first)
- STL export with companion text file of recommended settings

Effort: **~1-2 weeks** focused work. No new geometry math; just UI + data tables.

Phase 2.2 — 3MF profile bake for top printers

- 3MF writer that bundles STL + Bambu / Orca / Prusa preset
- 3 printer profiles initially (Bambu P1S, X1C, Prusa MK4)
- "Open in slicer" button (best-effort cross-platform)

Effort: **~1 week** on top of 2.1.

Phase 2.3 — Mold workflow (basic)

- New Output Mode dropdown
- Mold-design view with parting-plane slider
- Live 3D preview (Three.js) of mold halves
- Pour spout + auto-placed vents
- Two-STL export
- Operator instruction sheet PDF

Effort: **~3-4 weeks** focused work. Geometry math + UI + preview.

Phase 2.4 — Mold workflow (advanced)

- Registration features (pegs, dovetails)
- Identification embossing
- Validation panel with all §4.12 checks
- 3MF mold export with embedded rigid-print profile

Effort: **~2 weeks** on top of 2.3.

Phase 2.5 — Profile system expansion

- Profiles for remaining printers (§5.4–5.10)
- "Both" output mode
- Per-clinic profile override (admin-editable defaults)

Effort: **~1 week** for the printer profiles; admin editing is more.

Total spec effort estimate

~8-10 **focused weeks** for full v2 across all phases. Earliest valuable ship is 2.1 (~2 weeks) — the per-printer settings panel alone removes the most common operator confusion.

9. Dependencies & risks

RISK	MITIGATION
Bambu / Prusa change 3MF profile format	Bake formats are stable; pin tested versions; ship updates as needed
Boolean mesh operations on patient anatomy can produce non-manifold output	Use <code>pymeshlab</code> for boolean (more robust than <code>trimesh</code>); always run <code>is_watertight()</code> check post-op
Three.js performance on large meshes (chest-wall bolus = high triangle count)	Decimate mesh for preview only; keep full-resolution STL for export
Operator confusion between "trim base plate" workflow and "mold workflow"	Separate them in the UI; never combine in the same flow
Flexibolus formula or part numbers change	Cast-process info in §4.11 stays generic ("per Flexibolus product spec"); operator follows vendor doc
Embedded printer profiles get stale as slicer versions advance	Version each profile against a tested slicer release; show "Last tested: Bambu Studio 1.9" in UI
Operator picks wrong cast material in §4.1 by accident	All downstream defaults are <i>defaults</i> , not locks — operator can override

10. Out-of-scope items worth tracking

- **Direct LAN printing** (QuikBolus → Bambu MQTT → printer): separate spec
- **Auto-orient bolus to anatomy** from DICOM landmarks: separate spec
- **Sterilization / disinfection guidance** beyond "no autoclave": belongs in the operator print guide, not in QuikBolus itself
- **HDR cap bolus templates** (parametric library): separate Phase 3 spec
- **Tissue compensator / wedge generation**: distinct enough from bolus to be its own product, not a QuikBolus extension

Appendix A — Material reference

MATERIAL	DENSITY (G/CC)	CT NUMBER (HU)	USE FOR
Water (reference)	1.000	0	TPS assumption
TPU 95A (100% infill)	1.21	~+75	Direct-print bolus
TPU 95A (90% gyroid)	~1.09	~+45	Direct-print bolus, density-matched (within TG-176)
TPU 85A	1.13	~+60	Softer direct-print bolus
PETG	1.27	~+90	Rigid mold
PLA	1.24	~+85	Rigid mold (low-cost)
ABS	1.04	~+30	Hot-pour silicone mold
Flexibolus (cast)	~1.00	~0	Cast bolus — water-equivalent by design
Smooth-On Ecoflex 30	~1.07	~+35	Cast bolus (silicone)
Onyx (nylon + CF, Markforged)	1.20	~+70	Highly durable mold (production)

(HU values are typical; verify on your CT.)

Appendix B — Printer comparison matrix (quick reference)

PRINTER	BUILD VOL (MM)	ENCLOSURE	TPU FRIENDLY	BEST FIT
Bambu P1S	256 ³	Yes	✓✓✓	Default direct-print TPU
Bambu P1P	256 ³	No	✓✓	Direct-print TPU (drafts OK)
Bambu X1C	256 ³	Yes	✓✓✓	Direct-print TPU + AMS for rigid molds
Bambu A1	256 ³	No	✓✓	Smaller clinic budget
Bambu A1 mini	180 ³	No	✓	Small H&N patches only
Prusa MK4	250×210×220	No	✓✓✓	Direct-print TPU (proven workhorse)
Prusa XL	360×360×360	No	✓✓	Large chest-wall (single tool); multi-tool variant for PVA supports
Raise3D Pro3	300 ³	Yes	✓✓	Industrial production environments
Ultimaker S5	330×240×300	Yes	✓✓	PVA supports via BB extruder = great for molds
Markforged Mark Two	320×132×154	Yes	(Onyx)	Production-grade rigid molds, not TPU

Document control

VERSION	DATE	AUTHOR	CHANGE
0.1	2026-05-24	Radiant MPC / Claude	Initial draft from design conversation

Next step: review with QuikBolus owner, choose phases to commit to a release window, scope effort per phase, then turn this spec into engineering tasks. Until then this lives in `design-docs/` and is excluded from the public site via `.assetsignore`.